

BEYOND CORRECTNESS: EVALUATING SEMANTIC KNOWLEDGE IN CROSS-TABLE TRANSFER

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ABSTRACT

Semantic knowledge from feature names, documentation, codebooks, or knowledge graphs is often used in cross-table transfer to interpret heterogeneous schemas. Its contribution is commonly evaluated through ablations that compare intended semantic knowledge with altered alternatives, such as corrupted or shuffled semantics, or with omission-based controls. However, these controls can change more than the tested semantic components: altered conditions may degrade prediction, while omission can change the input or modeling setup. We therefore distinguish content sensitivity from predictive utility. Content sensitivity asks whether performance changes when semantic content is altered. Predictive utility asks whether the semantic component improves prediction relative to a reference that removes only that component while preserving the remaining information and modeling setup. Across real cross-table transfer and controlled experiments, the two measures can diverge substantially, often because altered conditions perform worse than the reference. Predictive utility also varies across references and increases when the tested information is harder to recover from other inputs. In an audit of 25 published comparisons across nine studies, only 1 of 18 explicit predictive-utility claims is paired with a control that meets all three criteria; 11 controls fail at least one criterion and 6 remain indeterminate. These results motivate claim-matched evaluation: what an ablation supports depends on what its control removes, alters, or preserves.

1 INTRODUCTION

Tabular datasets often describe related aspects of a common domain while organizing them under different schemas. Across datasets, corresponding variables may differ in their names, coding conventions, units, scales, or measurement protocols, even when they represent the same underlying concept. Predictive models typically rely on schema-specific input representations, making direct transfer difficult when the source and target schemas differ. Cross-table transfer often relies on semantic knowledge to interpret and relate heterogeneous schemas (Wang & Sun, 2022; Kim et al., 2024; Hegselmann et al., 2023). Such information can come from feature names, dataset documentation, codebooks, knowledge graphs, or domain rules (Pang et al., 2015; Li et al., 2025; Ruiz et al., 2023; Han et al., 2024), and can help interpret recorded values, identify corresponding variables, and introduce structured relationships useful for prediction.

In cross-table transfer, prior studies commonly assess semantic knowledge by comparing the intended knowledge with altered alternatives, such as corrupted or shuffled semantics, or with omission-based controls (Dinh et al., 2022; Hegselmann et al., 2023). Corrupted or shuffled alternatives may introduce misleading semantic information, degrading prediction and making the intended knowledge appear beneficial partly because the alternative performs worse. Omission-based controls can also be difficult to interpret because removing semantic knowledge can simultaneously remove other information available to the model, change how the remaining information is represented, or alter the transfer procedure (Hewitt & Liang, 2019; Elazar et al., 2021; Sturmfels et al., 2020). Consequently, a large semantic-ablation effect does not by itself establish a comparable improvement in prediction.

We therefore separate two evaluation questions that are often not clearly distinguished in semantic ablations. Content sensitivity asks whether predictive performance changes when the supplied se-

semantic content is altered while keeping fixed how that information is provided to and used by the model. Predictive utility asks whether the semantic component itself improves prediction relative to a suitable reference in which that component is removed. Evaluating predictive utility therefore requires a suitable reference that removes the evaluated component while preserving the remaining predictive information, its presentation, and the modeling setup. If no such reference can be constructed, the predictive utility of that component is not separately evaluable under the design; if several suitable references exist, utility should be examined across them. We apply this framework to semantic knowledge used for value interpretation, cross-schema correspondence, and variable relations.

Our experiments show that content sensitivity and predictive utility can lead to substantially different interpretations of the same semantic intervention. Large intended–altered gaps can be driven mainly by degradation under the altered condition, so strong content sensitivity need not imply a large predictive utility over a reference. Predictive utility can vary substantially across suitable references. In controlled experiments, it becomes larger when task-relevant information is harder for the reference to recover and smaller when that information is more readily recoverable. Across these manipulations, the intended condition changes comparatively little; much of the variation therefore reflects what the model can already obtain without the evaluated semantic knowledge. This distinction also matters in existing practice: our audit of 25 published semantic-ablation comparisons across nine studies finds that only a subset of reported controls clearly isolates the predictive utility of the semantic component being evaluated. Together, these findings show that the predictive utility of semantic knowledge depends not only on the knowledge itself, but also on what it is compared against and on the information already available to the model.

Contributions.

- **Conceptual.** We distinguish content sensitivity, which asks whether prediction depends on supplied semantic content, from predictive utility, which asks whether that knowledge improves prediction relative to a suitable reference.
- **Methodological.** We propose a claim-matched evaluation protocol that uses controls appropriate to the question and reports predictive utility only when a suitable reference can be constructed.
- **Empirical.** Across published ablations, real cross-table transfer, and controlled experiments, we show that content sensitivity can diverge from predictive utility. Predictive utility changes with the chosen reference and with how readily task-relevant information can be recovered from other inputs or labeled target data.

2 RELATED WORK

2.1 SEMANTIC KNOWLEDGE IN CROSS-TABLE TRANSFER

Cross-table methods such as TransTab and CARTE use semantic information to relate heterogeneous schemas: TransTab incorporates feature descriptions, while CARTE uses open-vocabulary representations (Wang & Sun, 2022; Kim et al., 2024). Related semantic tabular methods, such as TabLLM, serialize feature names and values as text (Hegselmann et al., 2023). More broadly, semantic or structured information in tabular learning can come from dataset documentation, schema-matching systems, knowledge graphs, language-model-generated features and rules, or task-derived behavioral constraints (Pang et al., 2015; Li et al., 2025; Ruiz et al., 2023; Han et al., 2024; Kim et al., 2025). These works primarily study how such information is obtained, represented, or used, leaving open whether a particular use improves prediction.

2.2 EVALUATING THE CONTRIBUTION OF SEMANTIC KNOWLEDGE

Across semantic tabular learning, ablations often compare standard semantic inputs with altered alternatives, such as shuffled or permuted inputs, or with omission-based controls, as in LIFT and TabLLM (Dinh et al., 2022; Hegselmann et al., 2023). Because these controls can change different aspects of the input, a performance gap may reflect more than the contribution of the tested semantic knowledge.

Related work on probing, feature importance, and information-removal interventions similarly emphasizes that conclusions depend on what a control changes and what it keeps fixed (Hewitt & Liang,

2019; Fisher et al., 2019; Elazar et al., 2021). Work on attribution baselines further shows that conclusions about a component’s contribution can depend on the chosen reference point (Sturmfels et al., 2020). Separately, transfer-learning studies show that the value of transferred information can diminish as labeled target support increases (He et al., 2019; Tripuraneni et al., 2020; Zoph et al., 2020; Nguyen et al., 2020).

We therefore evaluate semantic interventions using controls matched to the claim being tested. We distinguish sensitivity to semantic content from predictive utility relative to a suitable reference, and examine how utility varies across suitable references.

3 EVALUATING THE CONTRIBUTION OF SEMANTIC KNOWLEDGE

3.1 EVALUATION DESIGN

We evaluate each semantic component under matched conditions, holding the evaluation data, model, training procedure, and all other semantic components fixed. Our cross-table experiments use a labeled source table and a target table, with labeled target rows for adaptation and the remaining rows for evaluation. In the TabLLM reanalysis, semantic knowledge is instead varied within the model input.

We study three recurring roles of semantic knowledge. *Measurement* knowledge interprets coding, polarity, scales, units, and missing-value meanings; *correspondence* knowledge identifies compatible variables across schemas; and *relation* knowledge specifies structure among variables, such as derived values or directional constraints. These roles organize the test cases and are not treated as a full taxonomy.

For each component, the *intended condition* uses its prespecified form. We use *control* as a general term for a condition against which the intended condition is compared. An *altered control* changes the tested semantic content while preserving how it is used in prediction, whereas a *removal control* omits that content. A removal control serves as a *suitable reference* only if it satisfies the criteria in Section 3.2.

Let Q_c denote performance under condition c . All conditions use the same evaluation rows and procedure, and higher is better for every metric. We define

$$\text{Predictive utility} = Q_{\text{intended}} - Q_{\text{reference}}, \quad (1)$$

$$\text{Content sensitivity} = Q_{\text{intended}} - Q_{\text{altered}}. \quad (2)$$

Table 1: Condition construction for measurement, correspondence, and relation knowledge.

Knowledge	Intended	Altered	Reference
Measurement	Specified coding or numerical conversion	Altered coding or numerical conversion	Raw source values
Correspondence	Specified source–target pairing	Permuted source–target pairing	No shared pairing; separate source and target columns
Relation	Specified relation and direction	Reversed direction or random relation	No added value or constraint

For correspondence, the reference keeps the same number of columns but places source and target values in separate columns rather than a shared matched column. In the controlled benchmark, known schema transformations define the intended measurement and correspondence knowledge, while the synthetic outcome mechanism defines the intended relations. In real-data experiments, intended knowledge is derived from documentation or domain sources before evaluation and is not assumed to be ground-truth correct.

3.2 CONSTRUCTING A REFERENCE

A suitable reference must remove the evaluated semantic component while preserving the rest of the comparison. Simply removing the component may also change the input; for example, removing feature names can turn a key–value input into a values-only input (Figure 1).

Removing the evaluated component yields a candidate reference, which is suitable only if it satisfies three criteria: (i) *Removal*: it removes the evaluated semantic component; (ii) *Preservation*: preserves all other information available to the model and how that information is presented, including feature identity, input width, and format; (iii) *Setup*: leaves the model, learning procedure, and other modeling choices unchanged.

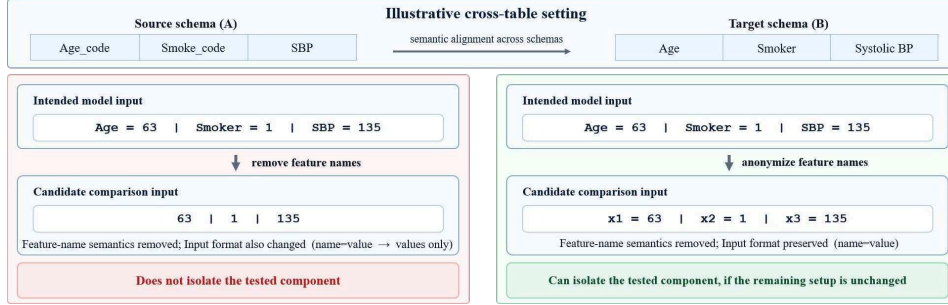


Figure 1: **Removing a semantic component does not necessarily yield a suitable reference.** Removing feature names entirely also changes the input format (left), whereas replacing them with anonymous identifiers preserves the key–value interface (right). The anonymous-identifier construction is illustrative rather than a proposed method.

For text inputs, changing the tested text can change tokens; this alone does not fail Preservation. For feature-name tests, Removal means that the original feature-name meaning is removed; we do not assume that every anonymous label has exactly the same effect on the model. In the published-ablation audit, we check Preservation by asking whether the other predictive values, feature identity, input width, and the rest of the input format are preserved, and check Setup by asking whether the model and learning procedure are unchanged Supp. §F.

Controls that fail these criteria may support broader comparisons but do not isolate predictive utility. Predictive utility is defined relative to a particular suitable reference, so different suitable references need not give the same value. If multiple references satisfy these criteria, we report reference sensitivity; if none does, predictive utility is not separately evaluable.

For a specified suitable reference, predictive utility is positive or negative when its 95% confidence interval lies entirely above or below zero; otherwise it is unresolved. Thus, *not separately evaluable* means that no suitable reference exists, whereas *unresolved* means that a suitable reference exists but the effect direction is not resolved. Resolution diagnostics are reported in Supp. §E.3.

In practice, claim-matched evaluation states the claim, selects a matching altered or removal control, and, for predictive utility, verifies Removal, Preservation, and Setup before reporting the contrast.

3.3 CONSTRUCTING SEMANTIC KNOWLEDGE

To evaluate the contribution of each role of semantic knowledge separately, we construct measurement, correspondence, and relation knowledge. Target labels and predictive performance are not used to create, select, or revise these specifications.

Measurement knowledge is obtained from dataset documentation, including codebooks, value labels, questionnaire wording, units, and measurement protocols. These sources determine coding, polarity, missing-value meanings, and documented unit conversions. We do not infer interpretations from observed values when documentation is missing or contradictory.

For correspondence knowledge, GPT-5.6 Sol proposes groups of related source and target variables from names, descriptions, units, value definitions, and dataset documentation, without access to row values or target labels. Groups may contain multiple variables from either schema and must cite the supplied documentation and pass the fixed validation checks described in the supplement.

For relation knowledge, candidates are constructed from dataset documentation and fixed before predictive evaluation. Each retained relation specifies the variables involved, an executable operation, and a direction when applicable. After the relation set is fixed, we review the literature only to

document support; the review is not used to add, remove, or revise relations. The supplement provides the prompts, schemas, validation rules, and execution records for the LLM-based construction step (Supp. §A.6).

4 EXPERIMENTS AND RESULTS

4.1 EXPERIMENTAL SETTINGS

Our quantitative evaluation spans three settings summarized in Table 2: the TabLLM reanalysis, real cross-table transfer, and the controlled benchmark. Clinical experiments use NHANES (NH), KNHANES (KN), and HRS (Johnson et al., 2013; Kweon et al., 2014; Sonnegga et al., 2014), and hydrological experiments use CAMELS-US and CAMELS-GB (Addor et al., 2017; Coxon et al., 2020). Clinical tasks include binary questionnaire endpoints and continuous laboratory or anthropometric endpoints. The TabLLM reanalysis examines a semantic ablation in an existing model, while the clinical, hydrological, and controlled experiments evaluate cross-table transfer directly.

Table 2: Summary of the experimental settings.

Setting	Data and task	Primary model	Metric
TabLLM reanalysis	Nine original-study datasets; released-code reanalysis with the name-masked reference	TabLLM (released)	AUROC
Clinical	NH–KN and pooled NH/KN–HRS; binary questionnaire and continuous examination endpoints	XGB	AUROC (binary); negative SD-normalized RMSE (continuous)
Hydrological	CAMELS-US to CAMELS-GB; 120 basins (CAMELS-120)	XGB	Negative log-RMSE
Controlled benchmark	Eight-variable NHANES panel with observed values and missingness retained; constructed schemas and synthetic targets	XGB	Negative nMSE

Note. Higher is better for all metrics. XGB is the primary cross-table learner; TransTab and CARTE are additionally evaluated in model-coverage analyses Supp. §E.2.2. For the controlled benchmark, $\text{nMSE} = \text{MSE}/\text{Var}(y)$. Full dataset, support, seed, learner, and inference settings are provided in Supp. §A.1.

Measurement knowledge transforms recorded values, correspondence knowledge matches source and target variables, and relation knowledge supplies derived values or directional constraints. Within each comparison, the learner and fitting procedure are otherwise unchanged.

For the controlled benchmark, we use real NHANES feature values while varying schema mismatch and prediction-task structure independently. Source and target schemas differ in coding polarity, numerical scale, missing-value coding, and column identifiers. Synthetic outcomes follow additive, pairwise, or sparse functions, so the schema differences and outcome mechanisms are fully specified.

For the TabLLM reanalysis, we use the released 11B T0/IA3 checkpoint with a name-masked reference that removes feature-name semantics while preserving feature identity and input format. Our released-code compatibility reproduction places 21 of 27 published means within .015 AUROC Supp. §B.1.

We use n_{adapt} for labeled TabLLM adaptation examples and n_{target} for labeled target examples in cross-table transfer. Conditions share the same split and adaptation or target samples, and the same realization in the controlled benchmark, so all contrasts are paired. Unless otherwise stated in the supplement, confidence intervals use 10,000 paired resamples over endpoint–seed pairs for clinical experiments, basins for hydrology, realizations for the controlled benchmark, and test rows for TabLLM. Prespecified analyses form the primary results; post-hoc analyses are explicitly labeled and are accompanied by the corresponding broader analysis when available Supp. §A.1.

We use these settings to test three questions in sequence: whether content sensitivity reflects predictive utility (4.2), whether utility varies across suitable references (4.3), and how utility changes with the information available to the reference (4.4).

4.2 CONTENT SENSITIVITY DOES NOT IDENTIFY PREDICTIVE UTILITY

A large intended–altered gap does not necessarily measure predictive utility because it can also reflect degradation under the altered condition. Content sensitivity can be decomposed as

$$\text{Content sensitivity} = \underbrace{(Q_{\text{intended}} - Q_{\text{reference}})}_{\text{predictive utility}} + (Q_{\text{reference}} - Q_{\text{altered}}). \quad (3)$$

Equation 3 is an identity; empirically, we ask how large the reference–altered term is and whether it explains most of the observed intended–altered gap. We apply this decomposition to TabLLM’s feature-name ablation, real-data relation transfer, and the controlled benchmark. In the relation experiments, switching from reversed to random directions changes only the altered condition, while the intended and reference conditions remain fixed. Predictive utility therefore remains unchanged, allowing us to examine directly how the altered condition affects content sensitivity. The controlled benchmark provides the analogous comparison when the intended relation is known exactly from the synthetic outcome design. Table 3 summarizes the decomposition.

Table 3: Decomposition of content sensitivity across the evaluated settings.

Setting	Altered condition	Content sensitivity	Predictive utility	Reference – Altered
<i>TabLLM feature-name ablation</i>				
9-dataset mean	<i>List Permuted Names</i>	+.112	+.088	+.024
Credit-g	<i>List Permuted Names</i>	+.066	−.003	+.069
Bank	<i>List Permuted Names</i>	−.032	+.089	−.121
<i>Real cross-table transfer: relation knowledge</i>				
NH–KN	Reversed	+.183	−.002	+.185
	Random	−.001	−.002	+.001
NH/KN–HRS	Reversed	+.266	+.002	+.264
	Random	+.041	+.002	+.039
CAMELS-120	Reversed	+.144	.000	+.144
	Random	+.002	.000	+.002
<i>Controlled benchmark</i>				
Additive XGB ($n_{\text{target}} = 32$)	Reversed	+.304	+.024	+.280
	Random	+.126	+.024	+.102

Note. For TabLLM, predictive utility uses the name-masked reference. Metrics are AUROC (TabLLM), negative SD-normalized RMSE (NH–KN and NH/KN–HRS), negative log-RMSE (CAMELS-120), and negative nMSE (controlled benchmark); higher is better. Reference – altered is the second term in Equation 3; positive values mean that the altered condition performs worse than the reference. Additional intervals and relation-control details are in Supp. Tables A6 and A9.

TabLLM shows that content sensitivity can either overstate or understate predictive utility. On Credit-g, content sensitivity is +.066, whereas predictive utility is −.003, because the altered condition is .069 lower than the reference. Bank shows the reverse case: the altered condition outperforms the reference by .121, so content sensitivity understates predictive utility (−.032 versus +.089).

The relation experiments show the same distinction. Replacing reversed with random directions sharply reduces content sensitivity in all three real-data settings while leaving predictive utility unchanged. The controlled benchmark shows the same pattern when the intended relation is known exactly from the synthetic outcome design. For NH–KN with random directions, content sensitivity (−.001) nearly matches predictive utility (−.002) because the altered condition performs similarly to the reference.

For the reversed controls, the near-zero utility point estimates show that almost the entire intended–altered gap comes from degradation under the altered condition rather than improvement over the reference. CAMELS-120 has a narrow 95% CI around zero ([−.001, +.001]), whereas the clinical intervals remain wide.

The same separation also appears with TransTab. At $n_{\text{target}} = 32$, 74–80% of the intended–altered gap is due to altered-condition degradation; at $n_{\text{target}} = 512$, this fraction rises to 97–100%. Additional CARTE results are more mixed, further indicating that the magnitude and direction of predictive utility also depend on the learner and how the semantic information is supplied Supp. §E.2.2. Across these settings, the intended–altered gap depends on how the altered condition is constructed and need not reflect predictive utility.

4.3 PREDICTIVE UTILITY CAN VARY ACROSS REFERENCES

We evaluate 24 name-masked references that remove the original feature-name meanings but assign generic identifiers differently; values and the remaining prompt structure are unchanged. The nine-dataset mean utility remains positive under all 24 references, but dataset-level point-estimate ranges cross zero for four of nine datasets (Figure 2).

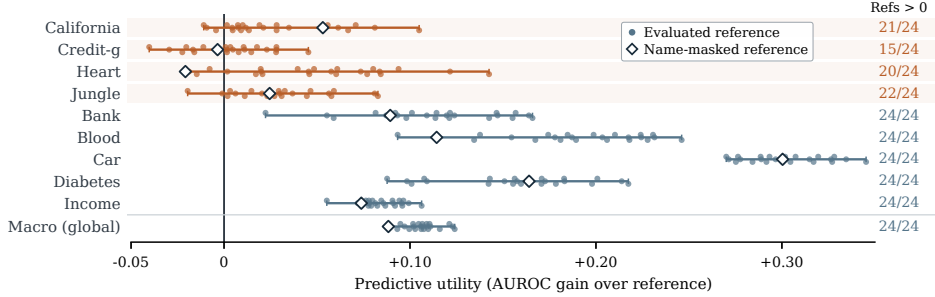


Figure 2: Reference sensitivity of TabLLM predictive utility. Points show the 24 evaluated identifier assignments, diamonds mark the primary name-masked reference, and horizontal lines show the point-estimate range; orange rows indicate ranges that cross zero. Counts at right show the number of positive estimates.

Across datasets, 79.2 – 99.7% of the variation across references remains after accounting for evaluation noise with paired resampling over shared test rows. Only for Jungle do simultaneous 95% confidence bands confirm a sign change, with one band entirely below zero and another entirely above zero. The other three crossings rest on point estimates only and remain unresolved. Reference choice clearly affects utility magnitude, while evidence that it changes the resolved sign is much more limited.

Reference-dependent variation remains when we restrict the analysis to the four datasets whose published TabLLM results we reproduce most closely. Similar variation also appears with alternative generic identifier labels, although zero crossings are less frequent. Satisfying the reference criteria allows predictive utility to be estimated, but different suitable references can still yield different values. We therefore recommend reporting reference sensitivity when multiple suitable references are available.

4.4 PREDICTIVE UTILITY DEPENDS ON WHAT THE REFERENCE CAN RECOVER

We next test whether predictive utility depends on how easily the reference can infer the tested information from other inputs or labeled target examples. We test this in two complementary ways: making that information harder to recover from observed inputs and easier to recover from labeled target examples. As a real-data reference point, measurement knowledge has positive utility across 10 clinical endpoints (+.197, 95% CI [+ .110, +.294]).

Utility increases when information is harder to recover from observed inputs. In the controlled benchmark, we test how measurement utility changes as source–target measurement mismatch increases, making the required transformation harder for the reference to infer. For relation knowledge, we instead remove constituent inputs, making the supplied relation value harder to recompute. Within each comparison, the intended and reference conditions receive the same observed inputs Supp. §D.1, §D.2.

Across both interventions, utility increases as the tested information becomes harder to recover. At the largest measurement mismatch and $n_{\text{target}} = 32$, XGB utility reaches +.259 to +.260 across the three target structures. Relation utility similarly rises from +.002 to +.242 as the constituent inputs are removed. This increase comes mainly from deterioration of the reference rather than improvement of the intended condition. In the pairwise XGB setting, intended nMSE changes only from .708 to .741, while reference nMSE worsens from .710 to .983.

Removing constituent inputs also changes the information available to the model. To test whether reconstructibility itself is related to utility, we instead add noise to the constituent inputs while keeping the number of input columns fixed. At each fixed noise level, lower reconstruction R^2 shows a modest association with higher utility (within-setting Spearman ρ between $1 - R^2$ and utility, averaged over eight settings at each nonzero dose: $+.165$ to $+.329$; Supp. §D.2). Because this relationship cannot be isolated from other input changes in the real-data relation experiments, we treat it as a modest association observed in the controlled benchmark rather than a mechanism established in real-data transfer.

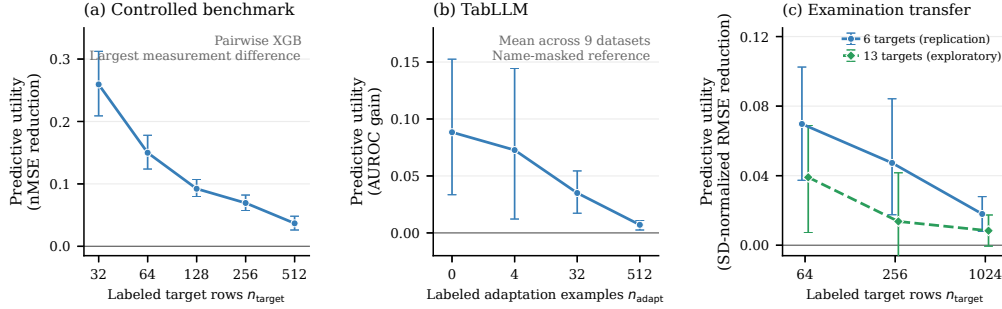


Figure 3: Predictive utility decreases as labeled target support increases. (a) Controlled measurement-knowledge utility for pairwise XGB at the largest tested measurement mismatch. (b) Mean TabLLM feature-name utility across nine datasets relative to the name-masked reference. (c) Correspondence-knowledge utility across a post hoc six-endpoint subset defined by whether the target itself is a matched variable, with the full 13-endpoint trajectory shown alongside it for comparison. Intervals are 95% CIs; metrics differ across panels and magnitudes are not directly comparable.

Utility decreases as labeled target support increases. Labeled target examples give the reference another way to recover task-relevant information. Accordingly, utility decreases as target support increases for the continuous clinical endpoints, in TabLLM, and the controlled benchmark (Figure 3).

Among the six examination endpoints for which the target itself is a matched variable, correspondence utility falls from $+.070$ to $+.018$ as target support increases. This attenuation is driven mainly by the reference catching up: intended SD-normalized RMSE improves from $.738$ to $.676$, while the reference improves from $.808$ to $.694$. The broader 13-endpoint analysis shows the same downward trend, although utility at $n_{\text{target}} = 256$ remains unresolved ($+.014$, 95% CI $[-.013, +.042]$).

The decline extends beyond continuous clinical endpoints. TabLLM utility falls from $+.088$ to $+.007$, a decrease of $.081$ (95% CI $[+.025, +.148]$), and utility is lower at $n_{\text{adapt}} = 512$ than at 0 under all 24 evaluated references. In the controlled benchmark, utility at the largest measurement mismatch falls from about $+.26$ to $+.03$ across the three prespecified target structures.

Together, these results show that utility is larger when the reference has fewer ways to recover the tested information. Making that information harder to recover increases utility, while labeled target examples allow the reference to catch up and reduce it. Across both manipulations, the intended condition changes comparatively little; most of the variation comes from reference performance. The measured utility also depends on the learner; additional capacity and learner-coverage analyses are reported in Supp. §E.2.1, §E.2.2.

4.5 IMPLICATIONS FOR PUBLISHED SEMANTIC ABLATIONS

Our experiments show that what a semantic-ablation gap can support depends on the control used in the comparison. We therefore ask whether published semantic ablations use controls that match their stated predictive claims. Using a fixed inclusion rule, we audit 25 comparisons from nine studies that explicitly alter or remove an identifiable semantic component Supp. §F. This is a bounded audit of claim-control alignment, not an estimate of its prevalence in the literature.

For each comparison, we record the claim and assess whether the reported control isolates the claimed effect of the tested component. For a predictive-utility claim, the control must remove

the tested semantic component while preserving the remaining information, input interface, model, and learning procedure. We classify the comparison as supported when all three requirements hold, unsupported when at least one clearly fails, and indeterminate when the published description is insufficient to establish them.

Table 4: Claim–control alignment in the published-ablation audit.

Claim type	Supported	Unsupported	Indeterminate
Content sensitivity ($n = 4$)	3	1	0
Predictive utility ($n = 18$)	1	11	6

**Note.* * A comparison may make both claim types, so the rows are not mutually exclusive. Counts include explicit claims only; for predictive utility, 18 comparisons make an explicit claim, five make no claim, and two have unclear scope.

Among the 18 explicit predictive-utility claims, only one is paired with a control that satisfies all three requirements; 11 fail at least one requirement, and six cannot be determined from the published description. Thus, most controls in this bounded sample do not by themselves isolate the predictive utility claimed for the tested semantic component. Importantly, this does not show that the reported semantic knowledge is unhelpful. It means only that the reported comparison does not isolate predictive utility under our criteria.

Because these judgments require manual coding, we also examined agreement between the two coders. Preservation had the lowest agreement in the initial coding ($\kappa = .265$; Supp. §F). We therefore replaced the broad Preservation judgment with six fixed checks, after which the same two coders independently re-coded all 25 comparisons before reviewing disagreements. Accordingly, the audit evaluates what each reported control can identify for the stated claim, rather than judging the overall correctness of the study or method.

5 CONCLUSION

Sensitivity to altered semantics does not by itself quantify the predictive contribution of semantic knowledge in cross-table transfer. Across our analyses, predictive utility depends on the reference, the information available without the tested knowledge, labeled target support, and the learner.

Altered controls remain informative: they can show whether prediction depends on the supplied semantic content, but they should not be interpreted as direct estimates of predictive utility. Our bounded audit shows that this distinction also matters in published practice, where many reported controls do not by themselves isolate the claimed predictive utility. These results support a claim-matched evaluation principle: controls should be chosen for the predictive claim they are intended to support. Predictive-utility claims instead require a suitable reference, and when several suitable references are available, variation across them should also be reported.

REPRODUCIBILITY STATEMENT

The supplementary material provides the experimental settings, inference procedures, analysis provenance, reference constructions, and robustness analyses required to reproduce the reported comparisons. The accompanying artifact contains code, frozen semantic records and reference libraries, analysis manifests, environment information, and mappings from reported tables and figures to their generating scripts and outputs. Public datasets can be obtained from their original sources; data requiring registration or license, including KNHANES and HRS, are not redistributed, and their access boundaries and reproducible derived outputs are documented in the supplement.

AI USE STATEMENT

Generative AI tools were used in this work in two roles. First, language models were used in parts of the semantic-knowledge construction process to propose candidates from schema metadata and dataset documentation. These steps did not use target labels or predictive performance, and the resulting semantic specifications were fixed before evaluation, as described in Section 3.3. Second, generative AI tools were used during research development to provide feedback on experimental

design and methodology and to assist with the interpretation of results. They were also used during manuscript preparation for drafting, editing, organization, and formatting. All experimental decisions, analyses, numerical results, references, and claims were reviewed and verified by the authors, who take responsibility for the final content of the work.

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